

Лабораторная работа №2

Тема: Настройка протокола STP (IEEE 802.1D)

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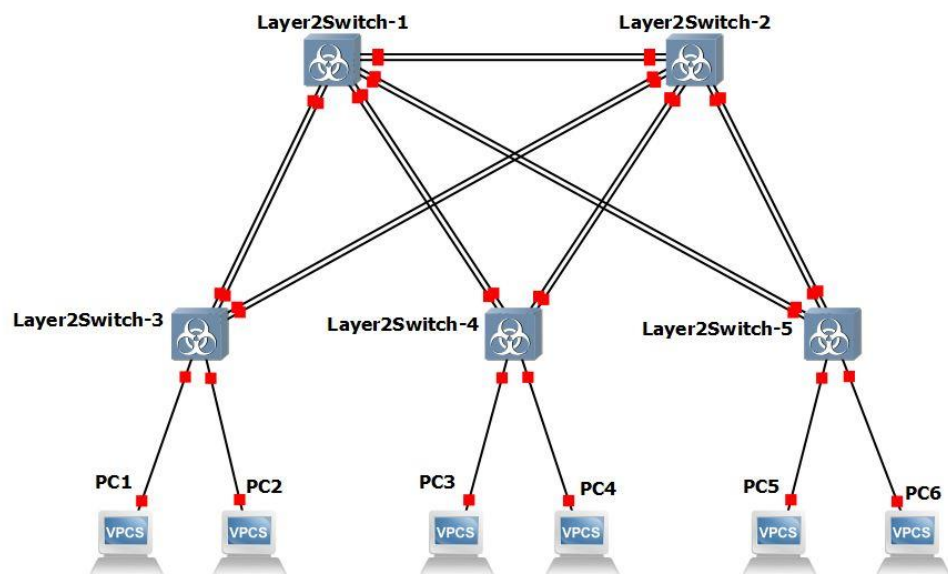


Рис. 1 - Схема, для которой необходимо настроить протокол STP.

IP адреса компьютеров:

ПК1: 192.168.16.1/24

ПК2: 192.168.16.2/24

ПК3: 192.168.16.3/24

ПК4: 192.168.16.4/24

ПК5: 192.168.16.5/24

ПК6: 192.168.16.6/24

После настройки протокола STP была выполнена проверка доступности всех ПК между собой:

```
PC1> ping 192.168.16.2

84 bytes from 192.168.16.2 icmp_seq=1 ttl=64 time=6.272 ms
84 bytes from 192.168.16.2 icmp_seq=2 ttl=64 time=7.702 ms
84 bytes from 192.168.16.2 icmp_seq=3 ttl=64 time=0.712 ms
84 bytes from 192.168.16.2 icmp_seq=4 ttl=64 time=5.957 ms
84 bytes from 192.168.16.2 icmp_seq=5 ttl=64 time=0.667 ms

PC1> ping 192.168.16.3

84 bytes from 192.168.16.3 icmp_seq=1 ttl=64 time=13.280 ms
84 bytes from 192.168.16.3 icmp_seq=2 ttl=64 time=4.644 ms
84 bytes from 192.168.16.3 icmp_seq=3 ttl=64 time=2.715 ms
84 bytes from 192.168.16.3 icmp_seq=4 ttl=64 time=6.498 ms
84 bytes from 192.168.16.3 icmp_seq=5 ttl=64 time=2.575 ms

PC1> ping 192.168.16.4

84 bytes from 192.168.16.4 icmp_seq=1 ttl=64 time=12.683 ms
84 bytes from 192.168.16.4 icmp_seq=2 ttl=64 time=8.654 ms
84 bytes from 192.168.16.4 icmp_seq=3 ttl=64 time=6.287 ms
84 bytes from 192.168.16.4 icmp_seq=4 ttl=64 time=8.335 ms
84 bytes from 192.168.16.4 icmp_seq=5 ttl=64 time=5.075 ms

PC1> ping 192.168.16.5

84 bytes from 192.168.16.5 icmp_seq=1 ttl=64 time=13.309 ms
84 bytes from 192.168.16.5 icmp_seq=2 ttl=64 time=8.375 ms
84 bytes from 192.168.16.5 icmp_seq=3 ttl=64 time=6.649 ms
84 bytes from 192.168.16.5 icmp_seq=4 ttl=64 time=1.751 ms
84 bytes from 192.168.16.5 icmp_seq=5 ttl=64 time=5.580 ms

PC1> ping 192.168.16.6

84 bytes from 192.168.16.6 icmp_seq=1 ttl=64 time=8.425 ms
84 bytes from 192.168.16.6 icmp_seq=2 ttl=64 time=6.801 ms
84 bytes from 192.168.16.6 icmp_seq=3 ttl=64 time=6.747 ms
84 bytes from 192.168.16.6 icmp_seq=4 ttl=64 time=7.564 ms
84 bytes from 192.168.16.6 icmp_seq=5 ttl=64 time=18.350 ms
```

Рис. 2 – проверка доступности ПК1 для других ПК.

```
PC2> ping 192.168.16.1

84 bytes from 192.168.16.1 icmp_seq=1 ttl=64 time=5.416 ms
84 bytes from 192.168.16.1 icmp_seq=2 ttl=64 time=1.705 ms
84 bytes from 192.168.16.1 icmp_seq=3 ttl=64 time=7.245 ms
84 bytes from 192.168.16.1 icmp_seq=4 ttl=64 time=7.846 ms
84 bytes from 192.168.16.1 icmp_seq=5 ttl=64 time=0.455 ms

PC2> ping 192.168.16.3

84 bytes from 192.168.16.3 icmp_seq=1 ttl=64 time=15.478 ms
84 bytes from 192.168.16.3 icmp_seq=2 ttl=64 time=8.234 ms
84 bytes from 192.168.16.3 icmp_seq=3 ttl=64 time=1.694 ms
84 bytes from 192.168.16.3 icmp_seq=4 ttl=64 time=1.136 ms
84 bytes from 192.168.16.3 icmp_seq=5 ttl=64 time=8.145 ms

PC2> ping 192.168.16.4

84 bytes from 192.168.16.4 icmp_seq=1 ttl=64 time=16.747 ms
84 bytes from 192.168.16.4 icmp_seq=2 ttl=64 time=15.652 ms
84 bytes from 192.168.16.4 icmp_seq=3 ttl=64 time=7.690 ms
84 bytes from 192.168.16.4 icmp_seq=4 ttl=64 time=6.820 ms
84 bytes from 192.168.16.4 icmp_seq=5 ttl=64 time=6.121 ms

PC2> ping 192.168.16.5

84 bytes from 192.168.16.5 icmp_seq=1 ttl=64 time=17.321 ms
84 bytes from 192.168.16.5 icmp_seq=2 ttl=64 time=5.369 ms
84 bytes from 192.168.16.5 icmp_seq=3 ttl=64 time=1.270 ms
84 bytes from 192.168.16.5 icmp_seq=4 ttl=64 time=1.759 ms
84 bytes from 192.168.16.5 icmp_seq=5 ttl=64 time=10.079 ms

PC2> ping 192.168.16.6

84 bytes from 192.168.16.6 icmp_seq=1 ttl=64 time=2.175 ms
84 bytes from 192.168.16.6 icmp_seq=2 ttl=64 time=4.108 ms
84 bytes from 192.168.16.6 icmp_seq=3 ttl=64 time=15.638 ms
84 bytes from 192.168.16.6 icmp_seq=4 ttl=64 time=8.070 ms
84 bytes from 192.168.16.6 icmp_seq=5 ttl=64 time=16.748 ms
```

Рис. 3 – проверка доступности ПК2 для других ПК.

```
PC3> ping 192.168.16.1

84 bytes from 192.168.16.1 icmp_seq=1 ttl=64 time=7.870 ms
84 bytes from 192.168.16.1 icmp_seq=2 ttl=64 time=5.547 ms
84 bytes from 192.168.16.1 icmp_seq=3 ttl=64 time=7.914 ms
84 bytes from 192.168.16.1 icmp_seq=4 ttl=64 time=7.796 ms
84 bytes from 192.168.16.1 icmp_seq=5 ttl=64 time=7.299 ms

PC3> ping 192.168.16.2

84 bytes from 192.168.16.2 icmp_seq=1 ttl=64 time=7.409 ms
84 bytes from 192.168.16.2 icmp_seq=2 ttl=64 time=6.975 ms
84 bytes from 192.168.16.2 icmp_seq=3 ttl=64 time=7.223 ms
84 bytes from 192.168.16.2 icmp_seq=4 ttl=64 time=3.499 ms
84 bytes from 192.168.16.2 icmp_seq=5 ttl=64 time=3.237 ms

PC3> ping 192.168.16.4

84 bytes from 192.168.16.4 icmp_seq=1 ttl=64 time=0.648 ms
84 bytes from 192.168.16.4 icmp_seq=2 ttl=64 time=1.802 ms
84 bytes from 192.168.16.4 icmp_seq=3 ttl=64 time=4.578 ms
84 bytes from 192.168.16.4 icmp_seq=4 ttl=64 time=0.640 ms
84 bytes from 192.168.16.4 icmp_seq=5 ttl=64 time=5.884 ms

PC3> ping 192.168.16.5

84 bytes from 192.168.16.5 icmp_seq=1 ttl=64 time=20.484 ms
84 bytes from 192.168.16.5 icmp_seq=2 ttl=64 time=2.545 ms
84 bytes from 192.168.16.5 icmp_seq=3 ttl=64 time=6.780 ms
84 bytes from 192.168.16.5 icmp_seq=4 ttl=64 time=7.373 ms
84 bytes from 192.168.16.5 icmp_seq=5 ttl=64 time=8.549 ms

PC3> ping 192.168.16.6

84 bytes from 192.168.16.6 icmp_seq=1 ttl=64 time=9.350 ms
84 bytes from 192.168.16.6 icmp_seq=2 ttl=64 time=4.567 ms
84 bytes from 192.168.16.6 icmp_seq=3 ttl=64 time=5.981 ms
84 bytes from 192.168.16.6 icmp_seq=4 ttl=64 time=6.047 ms
84 bytes from 192.168.16.6 icmp_seq=5 ttl=64 time=8.966 ms
```

Рис. 4 – проверка доступности ПК3 для других ПК.

```
PC4> ping 192.168.16.1

84 bytes from 192.168.16.1 icmp_seq=1 ttl=64 time=14.809 ms
84 bytes from 192.168.16.1 icmp_seq=2 ttl=64 time=6.838 ms
84 bytes from 192.168.16.1 icmp_seq=3 ttl=64 time=6.345 ms
84 bytes from 192.168.16.1 icmp_seq=4 ttl=64 time=7.718 ms
84 bytes from 192.168.16.1 icmp_seq=5 ttl=64 time=7.036 ms

PC4> ping 192.168.16.2

84 bytes from 192.168.16.2 icmp_seq=1 ttl=64 time=15.107 ms
84 bytes from 192.168.16.2 icmp_seq=2 ttl=64 time=6.222 ms
84 bytes from 192.168.16.2 icmp_seq=3 ttl=64 time=15.529 ms
84 bytes from 192.168.16.2 icmp_seq=4 ttl=64 time=7.617 ms
84 bytes from 192.168.16.2 icmp_seq=5 ttl=64 time=7.884 ms

PC4> ping 192.168.16.3

84 bytes from 192.168.16.3 icmp_seq=1 ttl=64 time=5.552 ms
84 bytes from 192.168.16.3 icmp_seq=2 ttl=64 time=3.012 ms
84 bytes from 192.168.16.3 icmp_seq=3 ttl=64 time=4.590 ms
84 bytes from 192.168.16.3 icmp_seq=4 ttl=64 time=7.183 ms
84 bytes from 192.168.16.3 icmp_seq=5 ttl=64 time=7.466 ms

PC4> ping 192.168.16.5

84 bytes from 192.168.16.5 icmp_seq=1 ttl=64 time=7.997 ms
84 bytes from 192.168.16.5 icmp_seq=2 ttl=64 time=1.268 ms
^[[A84 bytes from 192.168.16.5 icmp_seq=3 ttl=64 time=1.961 ms
84 bytes from 192.168.16.5 icmp_seq=4 ttl=64 time=1.785 ms
84 bytes from 192.168.16.5 icmp_seq=5 ttl=64 time=4.944 ms

PC4> ping 192.168.16.6

84 bytes from 192.168.16.6 icmp_seq=1 ttl=64 time=5.365 ms
84 bytes from 192.168.16.6 icmp_seq=2 ttl=64 time=12.720 ms
84 bytes from 192.168.16.6 icmp_seq=3 ttl=64 time=10.796 ms
84 bytes from 192.168.16.6 icmp_seq=4 ttl=64 time=6.949 ms
84 bytes from 192.168.16.6 icmp_seq=5 ttl=64 time=4.188 ms
```

Рис. 5 – проверка доступности ПК4 для других ПК.

```
PC5> ping 192.168.16.1

84 bytes from 192.168.16.1 icmp_seq=1 ttl=64 time=5.360 ms
84 bytes from 192.168.16.1 icmp_seq=2 ttl=64 time=1.677 ms
84 bytes from 192.168.16.1 icmp_seq=3 ttl=64 time=4.604 ms
84 bytes from 192.168.16.1 icmp_seq=4 ttl=64 time=7.181 ms
84 bytes from 192.168.16.1 icmp_seq=5 ttl=64 time=7.754 ms

PC5> ping 192.168.16.2

84 bytes from 192.168.16.2 icmp_seq=1 ttl=64 time=15.130 ms
84 bytes from 192.168.16.2 icmp_seq=2 ttl=64 time=1.474 ms
84 bytes from 192.168.16.2 icmp_seq=3 ttl=64 time=15.796 ms
84 bytes from 192.168.16.2 icmp_seq=4 ttl=64 time=7.674 ms
84 bytes from 192.168.16.2 icmp_seq=5 ttl=64 time=7.065 ms

PC5> ping 192.168.16.3

84 bytes from 192.168.16.3 icmp_seq=1 ttl=64 time=2.151 ms
84 bytes from 192.168.16.3 icmp_seq=2 ttl=64 time=1.661 ms
84 bytes from 192.168.16.3 icmp_seq=3 ttl=64 time=6.493 ms
84 bytes from 192.168.16.3 icmp_seq=4 ttl=64 time=5.756 ms
84 bytes from 192.168.16.3 icmp_seq=5 ttl=64 time=7.172 ms

PC5> ping 192.168.16.4

84 bytes from 192.168.16.4 icmp_seq=1 ttl=64 time=9.389 ms
84 bytes from 192.168.16.4 icmp_seq=2 ttl=64 time=4.637 ms
84 bytes from 192.168.16.4 icmp_seq=3 ttl=64 time=8.017 ms
84 bytes from 192.168.16.4 icmp_seq=4 ttl=64 time=6.613 ms
84 bytes from 192.168.16.4 icmp_seq=5 ttl=64 time=14.865 ms

PC5> ping 192.168.16.6

84 bytes from 192.168.16.6 icmp_seq=1 ttl=64 time=0.464 ms
84 bytes from 192.168.16.6 icmp_seq=2 ttl=64 time=3.401 ms
84 bytes from 192.168.16.6 icmp_seq=3 ttl=64 time=0.697 ms
84 bytes from 192.168.16.6 icmp_seq=4 ttl=64 time=1.640 ms
84 bytes from 192.168.16.6 icmp_seq=5 ttl=64 time=3.409 ms
```

Рис. 6 – проверка доступности ПК5 для других ПК.

```
PC6> ping 192.168.16.1

84 bytes from 192.168.16.1 icmp_seq=1 ttl=64 time=2.008 ms
84 bytes from 192.168.16.1 icmp_seq=2 ttl=64 time=7.735 ms
84 bytes from 192.168.16.1 icmp_seq=3 ttl=64 time=7.276 ms
84 bytes from 192.168.16.1 icmp_seq=4 ttl=64 time=7.585 ms
84 bytes from 192.168.16.1 icmp_seq=5 ttl=64 time=4.823 ms

PC6> ping 192.168.16.2

84 bytes from 192.168.16.2 icmp_seq=1 ttl=64 time=9.466 ms
84 bytes from 192.168.16.2 icmp_seq=2 ttl=64 time=8.790 ms
84 bytes from 192.168.16.2 icmp_seq=3 ttl=64 time=7.197 ms
84 bytes from 192.168.16.2 icmp_seq=4 ttl=64 time=7.550 ms
84 bytes from 192.168.16.2 icmp_seq=5 ttl=64 time=15.741 ms

PC6> ping 192.168.16.3

84 bytes from 192.168.16.3 icmp_seq=1 ttl=64 time=3.259 ms
84 bytes from 192.168.16.3 icmp_seq=2 ttl=64 time=2.595 ms
84 bytes from 192.168.16.3 icmp_seq=3 ttl=64 time=4.340 ms
84 bytes from 192.168.16.3 icmp_seq=4 ttl=64 time=1.689 ms
84 bytes from 192.168.16.3 icmp_seq=5 ttl=64 time=5.389 ms

PC6> ping 192.168.16.4

84 bytes from 192.168.16.4 icmp_seq=1 ttl=64 time=9.390 ms
84 bytes from 192.168.16.4 icmp_seq=2 ttl=64 time=7.743 ms
84 bytes from 192.168.16.4 icmp_seq=3 ttl=64 time=5.625 ms
84 bytes from 192.168.16.4 icmp_seq=4 ttl=64 time=14.668 ms
84 bytes from 192.168.16.4 icmp_seq=5 ttl=64 time=9.389 ms

PC6> ping 192.168.16.5

84 bytes from 192.168.16.5 icmp_seq=1 ttl=64 time=0.599 ms
84 bytes from 192.168.16.5 icmp_seq=2 ttl=64 time=4.376 ms
84 bytes from 192.168.16.5 icmp_seq=3 ttl=64 time=2.789 ms
84 bytes from 192.168.16.5 icmp_seq=4 ttl=64 time=2.503 ms
84 bytes from 192.168.16.5 icmp_seq=5 ttl=64 time=2.330 ms
```

Рис. 7 – проверка доступности ПК6 для других ПК.

Схема с отмеченными BID коммутаторов и стоимостями всех маршрутов

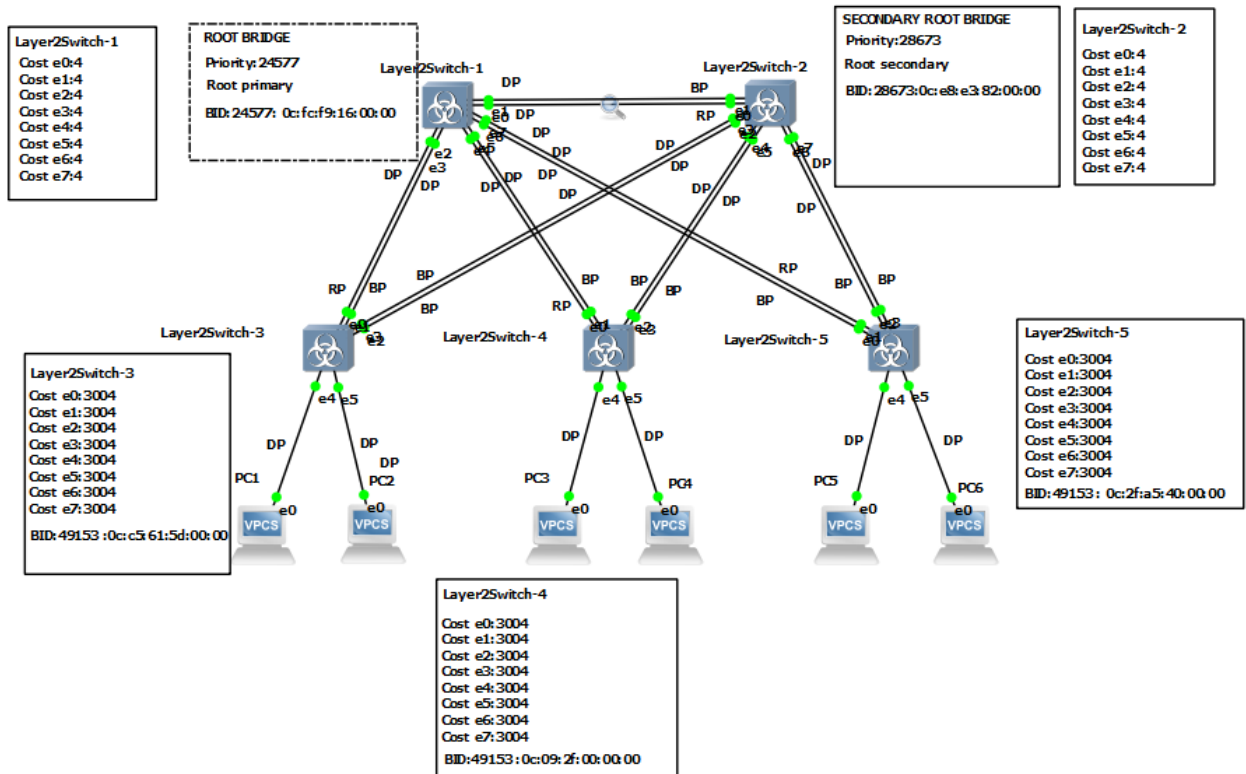


Рис. 8 – Схема с отмеченными BID коммутаторов и указанными стоимостями маршрутов.

RP – root port

DP – designated port

BP – blocked port

Отправка пакетов hello корневым коммутатором

Захват из - [Layer2Switch-1 Ethernet0 to Layer2Switch-2 Ethernet0]

Файл Редактирование Просмотр Захват Анализ Статистика Телефония Беспроводной Инструменты Помощь

Применить дисплейный фильтр ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:fc:f9:16:00:00	0c:fc:f9:16:00:00	LOOP	60	Reply
2	0.062975	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 24576/1/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
3	0.341772	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
4	0.344362	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
5	0.345665	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
6	1.062829	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 24576/1/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
7	1.306781	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
8	1.308794	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
9	1.309006	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
10	2.062675	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 24576/1/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
11	2.341467	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
12	2.344065	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
13	2.345349	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
14	3.062515	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 24576/1/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
15	3.306488	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
16	3.308493	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
17	3.308748	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
18	4.062384	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 24576/1/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
19	4.341116	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
20	4.343766	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
21	4.345022	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
22	5.062242	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 24576/1/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
23	5.306205	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
24	5.308194	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
25	5.308402	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
26	6.062296	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 24576/1/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
27	6.340774	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
28	6.343454	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
29	6.344745	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
30	7.061917	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 24576/1/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
31	7.305879	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
32	7.307889	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
33	7.308142	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
34	8.061762	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 24576/1/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
35	8.340531	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
36	8.343162	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
37	8.344447	0c:e8:e3:82:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:e8:e3:82:00:00 Cost = 0 Port = 0x8001
38	9.061609	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 24576/1/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
39	9.305586	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/100/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
40	9.307583	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/200/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
41	9.307835	0c:fc:f9:16:00:00	Spanning-tree-(for...	STP	60	Conf. Root = 32768/300/0c:fc:f9:16:00:00 Cost = 0 Port = 0x8001
42	9.430272	0c:e8:e3:82:00:00	0c:e8:e3:82:00:00	LOOP	60	Reply
43	9.998472	0c:fc:f9:16:00:00	0c:fc:f9:16:00:00	LOOP	60	Reply

Рис. 9 – Отправка пакетов hello от корневого коммутатора.

Изменение стоимости маршрута на коммутаторе Layer2Switch-2

Изменение стоимости маршрута выполняется командой `spanning-tree cost <value>`, причем команда должна быть прописана для определённого порта.

```

Layer2Switch-2(config)#int range Gi0/0
Layer2Switch-2(config-if-range)#spanning-tree cost 7
Layer2Switch-2(config-if-range)#exit
Layer2Switch-2(config)#^Z
Layer2Switch-2#
*Aug 30 05:30:36.131: %SYS-5-CONFIG_I: Configured from console by console
Layer2Switch-2#sh sp

VLAN0001
  Spanning tree enabled protocol ieee
    Root ID      Priority    24577
                Address      0cfc.f916.0000
                Cost        4
                Port        2 (GigabitEthernet0/1)
                Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID    Priority    28673 (priority 28672 sys-id-ext 1)
                Address      0ce8.e382.0000
                Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
                Aging Time   15 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Altn BLK 7        128.1    Shr

```

Измененная топология:

